

SUPSI

Presentazione Progetto Image Analysis

Studente/i	Relatore	Correlatore
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Corso di laurea	Modulo / Codice Progetto	Anno
Ingegneria Informatica	Progetto di Semestre M-P6202	2025-2026
Committente	Data	
MeDiTech, UniCt	20/05/2026	

Contesto e Motivazione

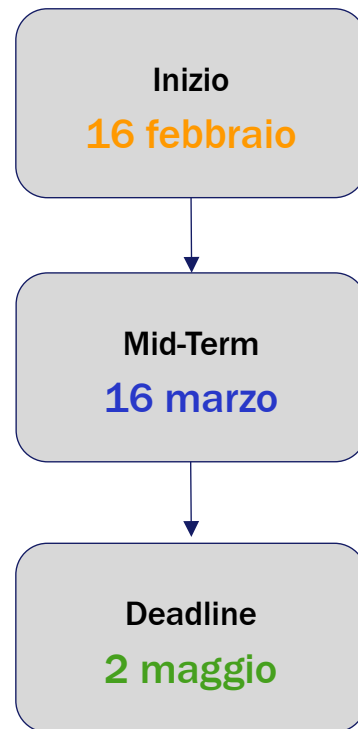
Progetto commissionato da uniCT e MeDiTech

Parte 1:

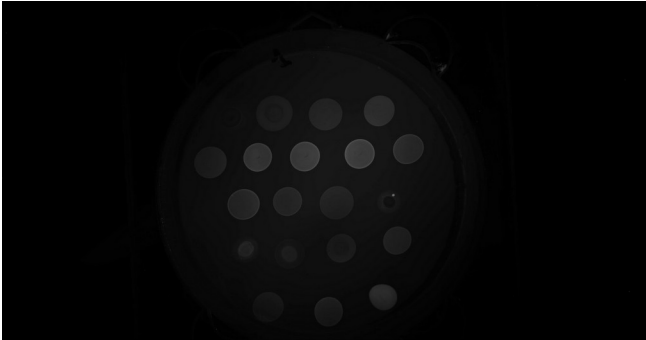
Pipeline automatica per confronto immagini spettrali di pozzetti per l'analisi dei livelli di espressione genica di biomarcatori target

Parte 2:

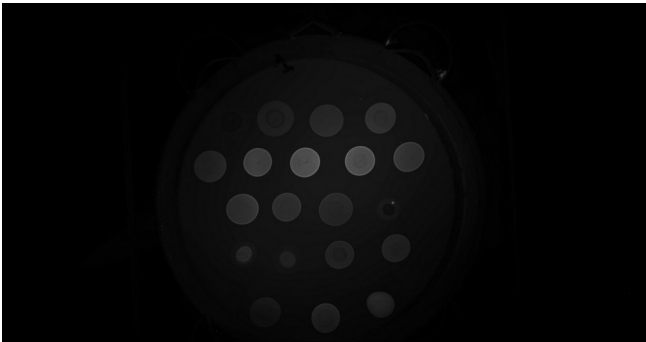
Applicazione desktop con interfaccia grafica
Deployment, testing, documentazione



Input



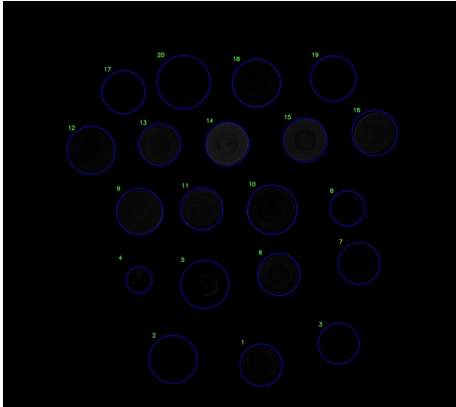
prima.avi



dopo.avi



Output

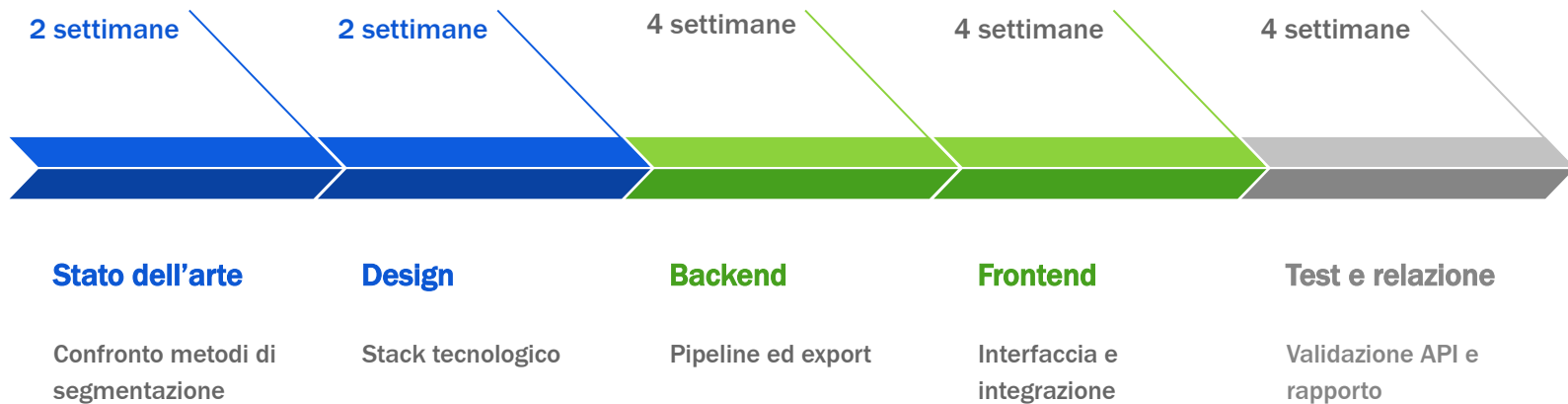


diff.jpg



analysis_export.csv

Project design



Analisi dei requisiti

Backend

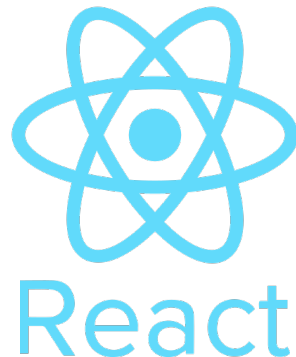
- 1) Individuazione automatica delle ROI
- 2) Allineamento video
- 3) Calcolo differenziale pixel-pixel
- 4) Esportazione analisi come CSV



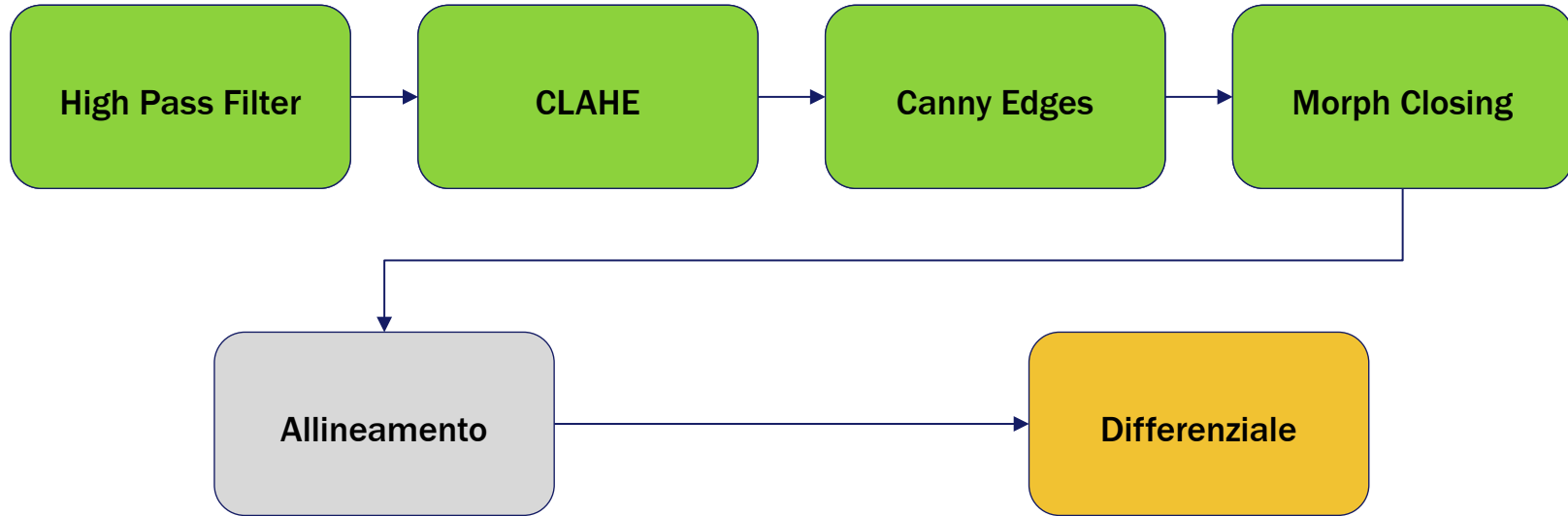
Analisi dei requisiti (2)

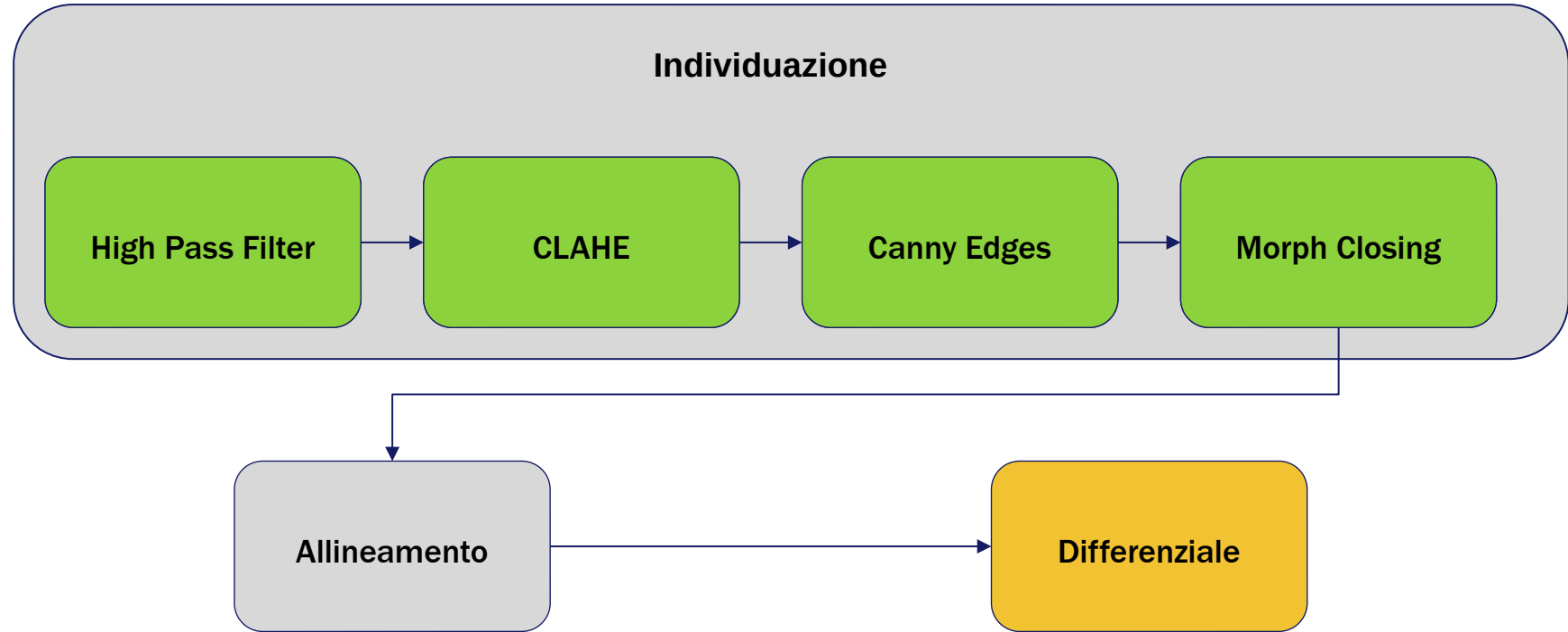
Frontend

- 1) Selezione video
- 2) Differenziale per ciascuna frequenza
- 3) Visualizzazione dettaglio singole ROI
- 4) Modifica parametri e per riapplicazione pipeline



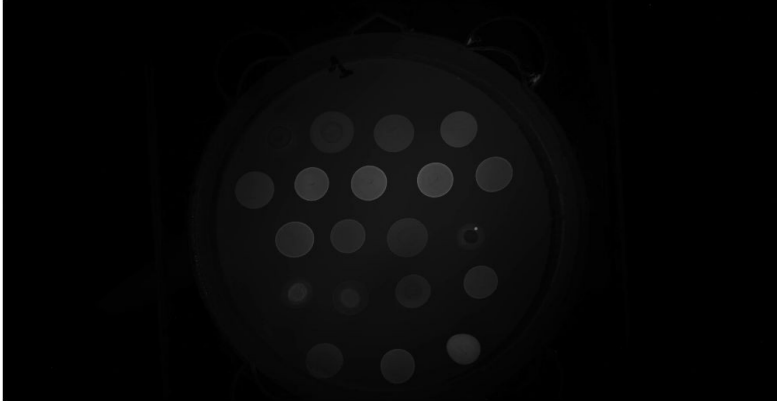
Overview Pipeline





Pipeline: Individuazione

Original



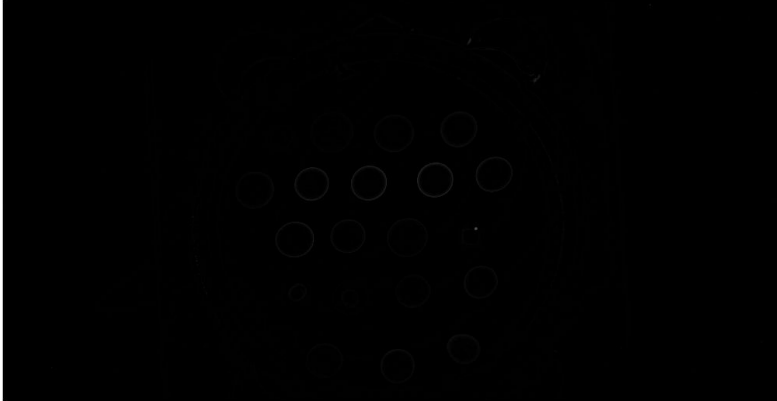
Objects



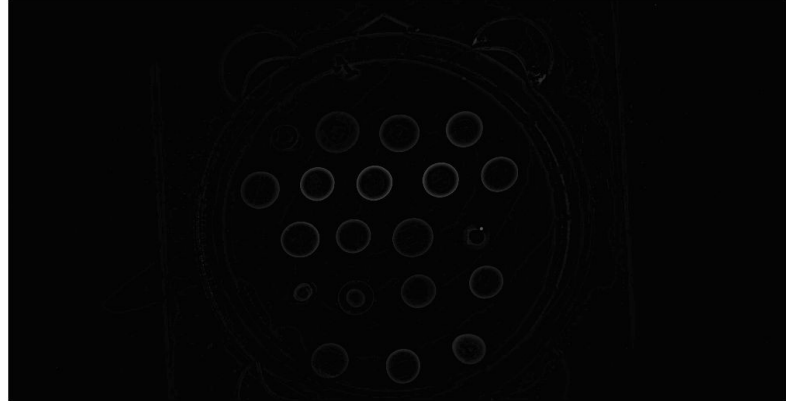
High Pass Filter (HPF)

Pipeline: Individuazione (2)

Objects



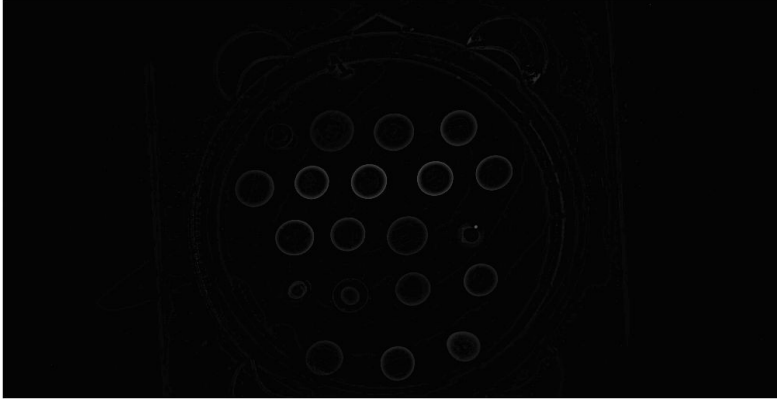
CLAHE



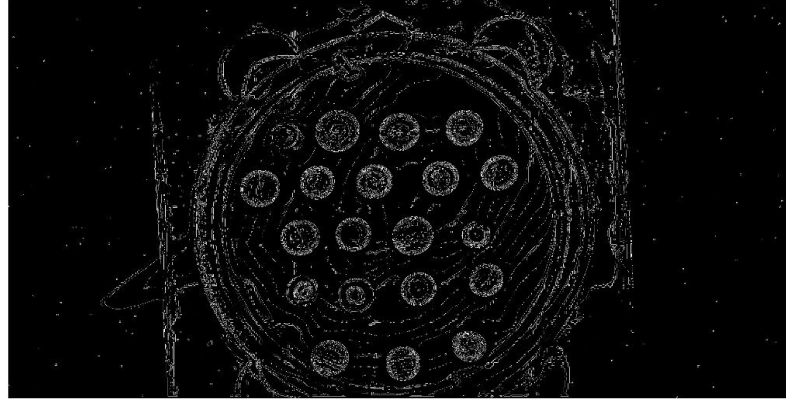
Contrast Limited Adaptive Histogram Equalization (CLAHE)

Pipeline: Individuazione (3)

CLAHE



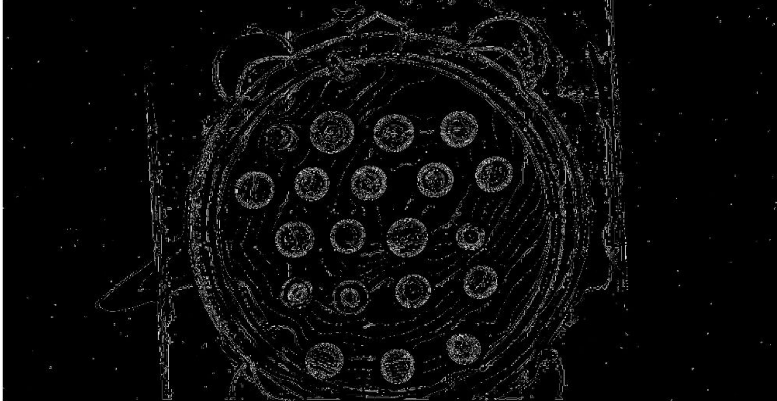
Canny Edges



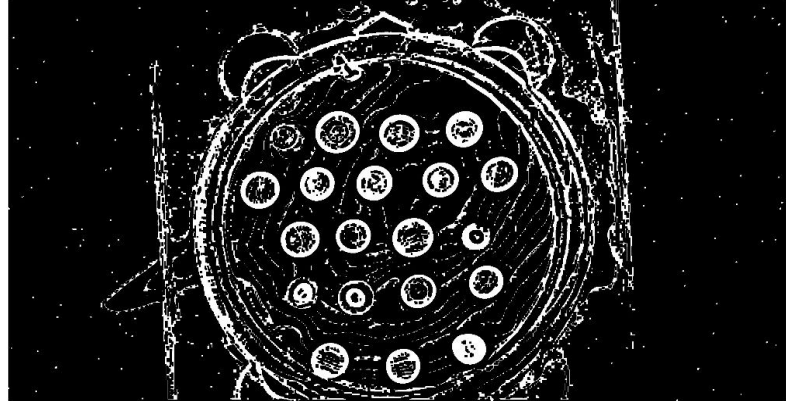
Canny Edges

Pipeline: Individuazione (4)

Canny Edges



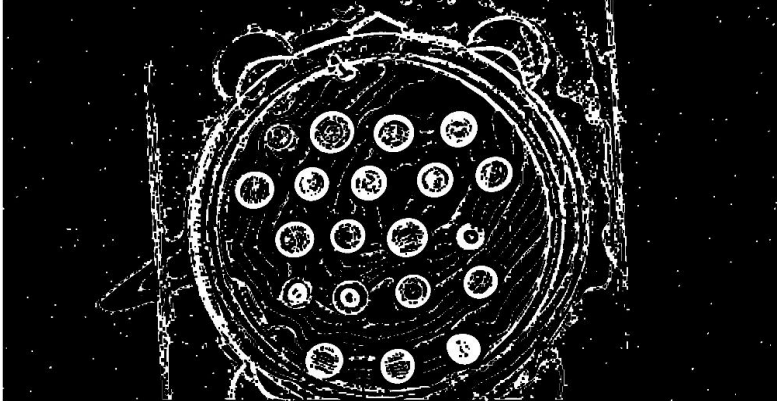
Morph Closing



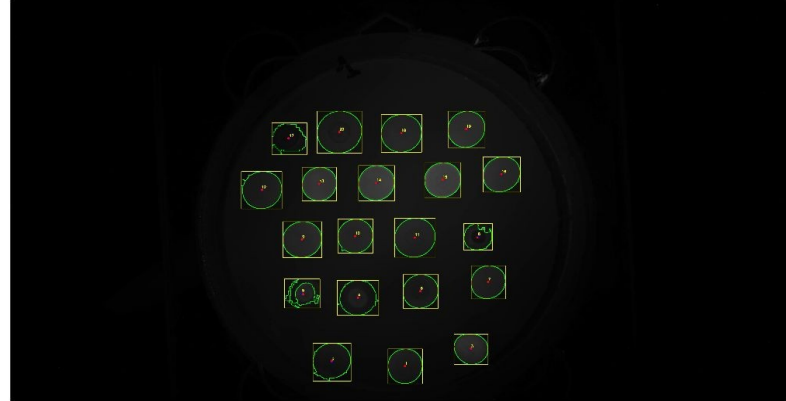
Morph Closing

Pipeline: Individuazione (5)

Morph Closing



Valid Contours



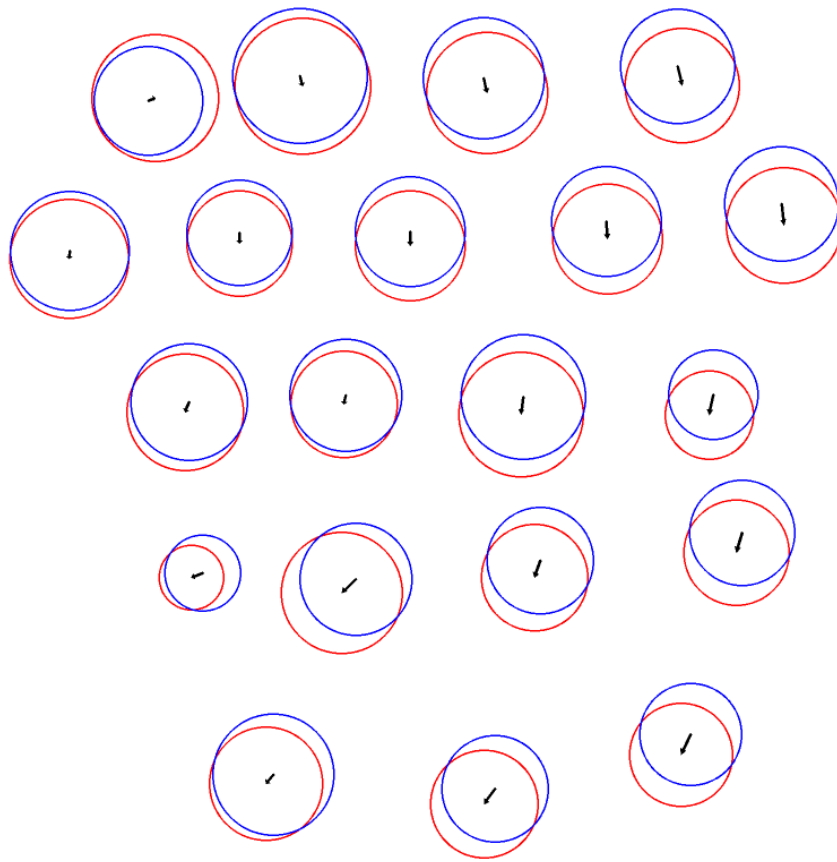
Contours

Pipeline: Allineamento

Spostiamo i contorni dell'analisi dopo sopra quelli dell'analisi prima

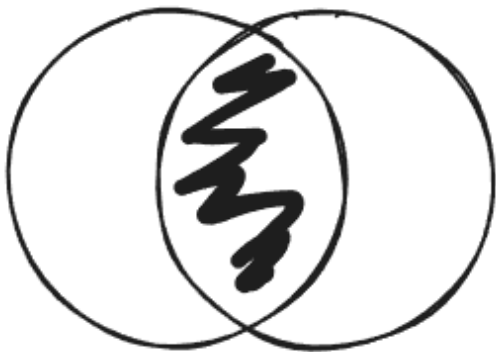
Rosso: analisi prima

Blu: analisi dopo

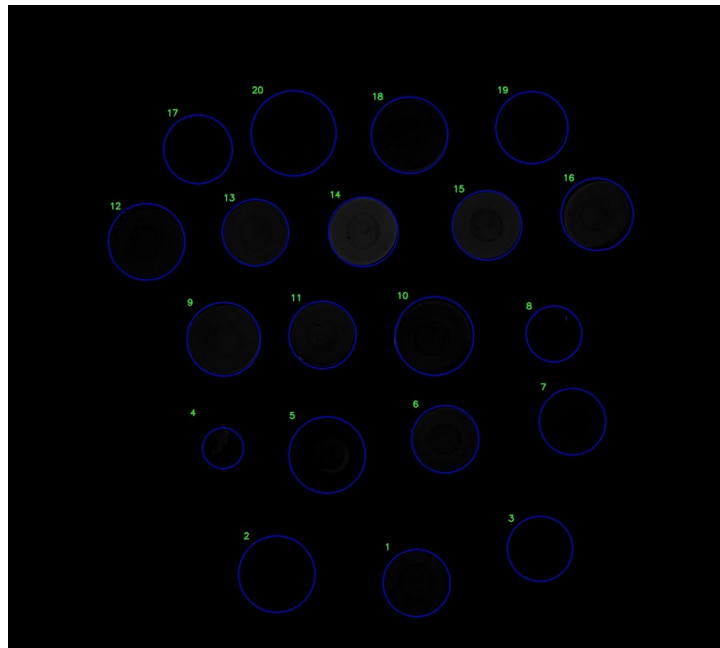


Pipeline: Differenziale

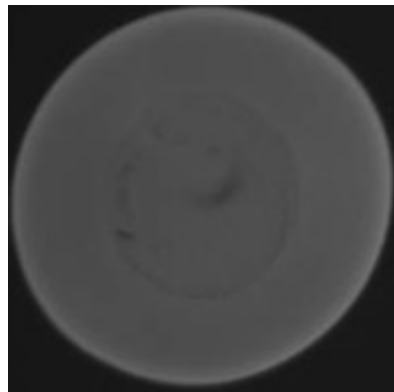
Prendiamo l'area comune e sottraiamo
il valore d'intensità pixel per pixel



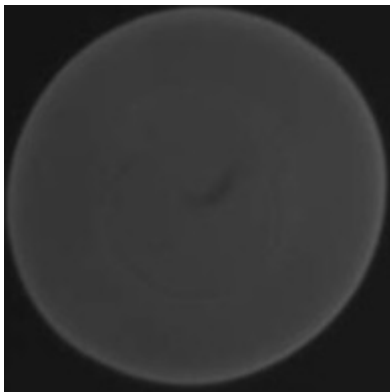
(Esempio esagerato)



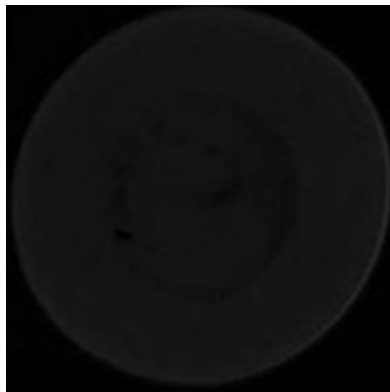
Risultati



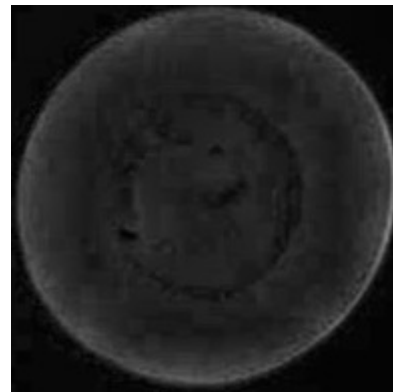
**ROI 14
Prima**



**ROI 14
Dopo**

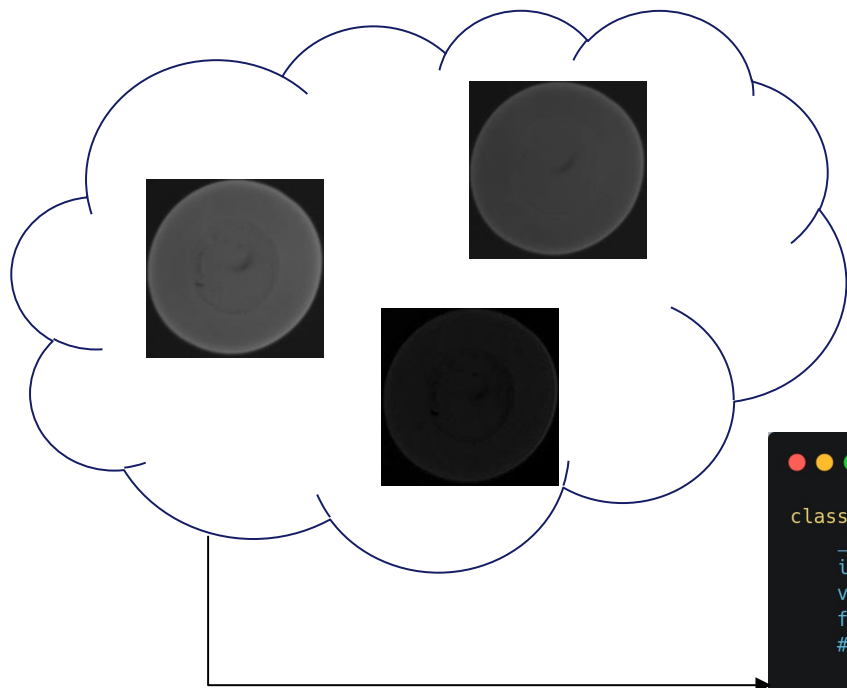


diff.



diff. + CLAHE

Risultati (2)



Modellizzazione di un problema
reale in **programmazione a
oggetti** con persistenza su db
relazionale



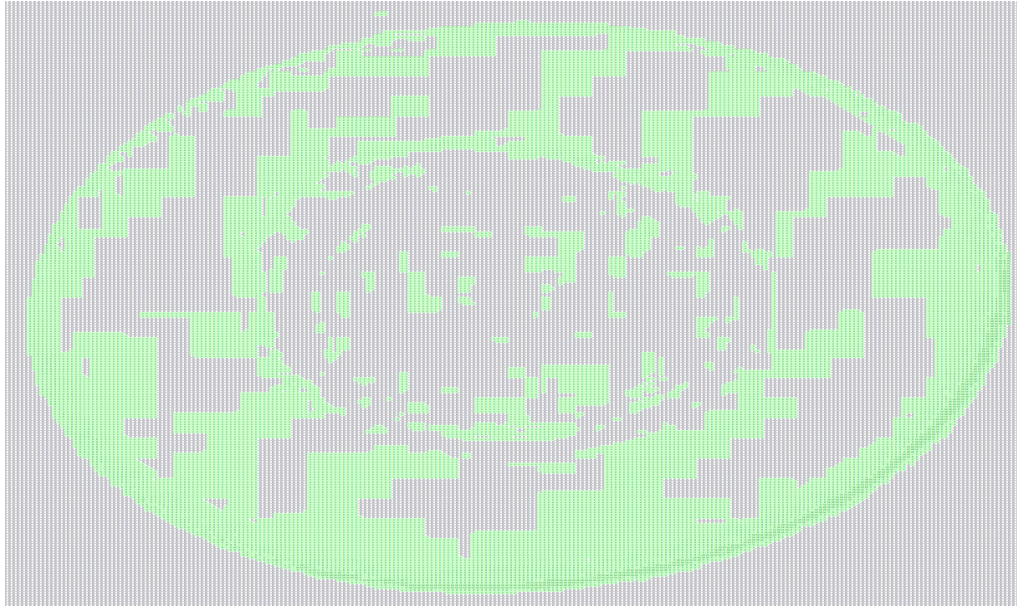
```
class Roi(SQLModel, table=True):  
    __tablename__ = "roi"  
    idx: int  
    video_path: str  
    fase: Analisi = Field(default=None, sa_column=Column(SAEnum(Analisi)))  
    # ...
```

Esportazione CSV

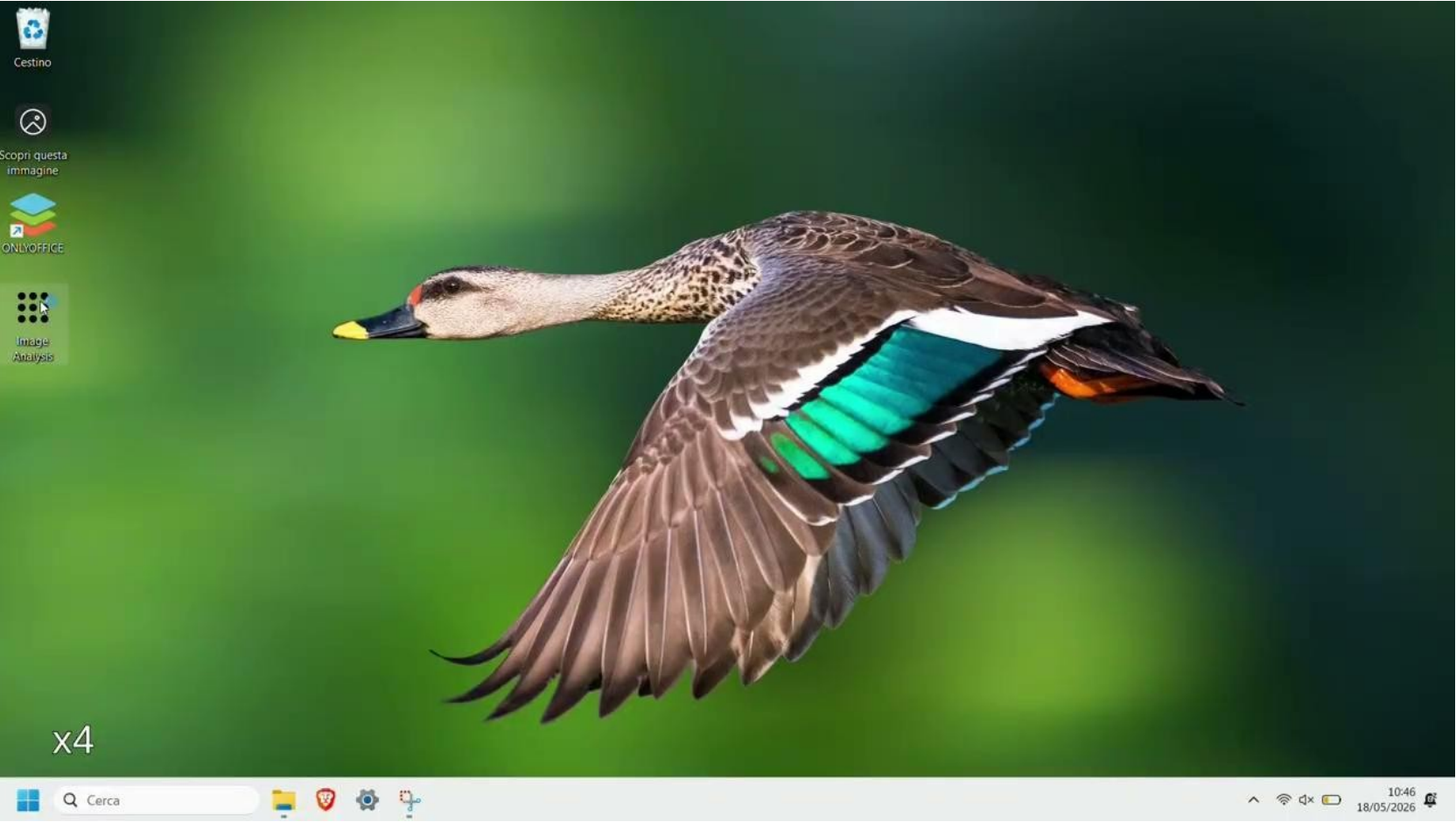
	FREQ_420	FREQ_421	frequenza
ROI_0	1.49781916808	1.5603426559	...
ROI_1	-0.1079808468	-0.0511343631	...
roi	...	...	intensità media diff.

roi_analysis_export.csv

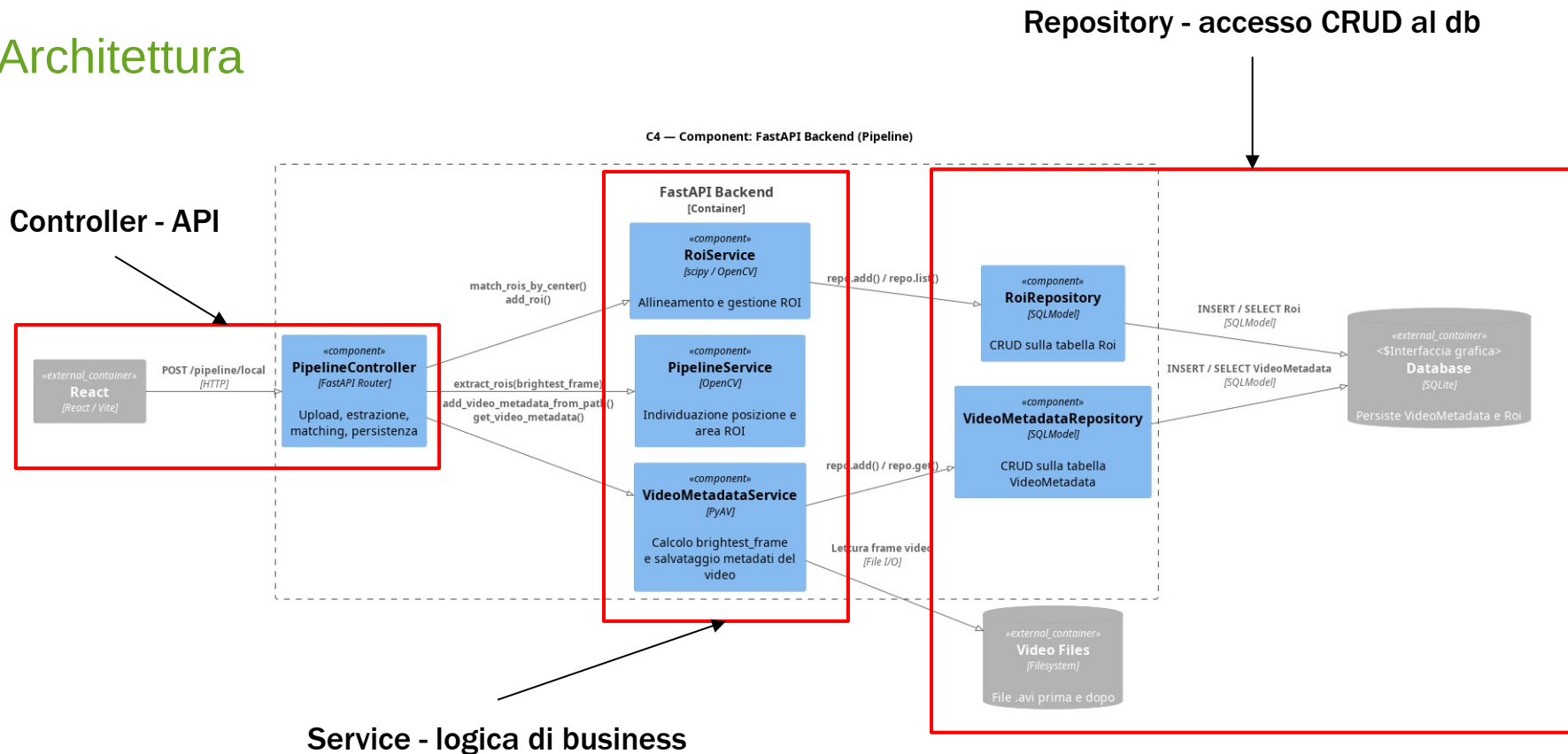
Esportazione CSV (2)



roi_analysis_export_pixels.csv



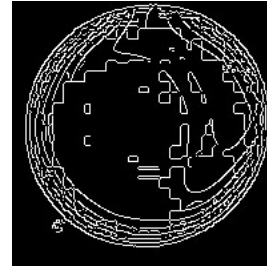
Architettura



Test

Testati tutti e 21 gli endpoint: output in immagini per ulteriore verifica manuale

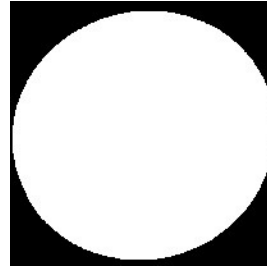
Test step pipeline



Test

Testati tutti e 21 gli endpoint: output in immagini per ulteriore verifica manuale

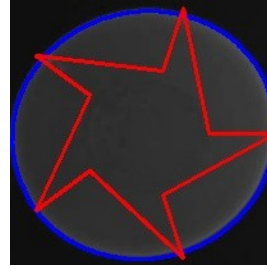
Test area comune



Test

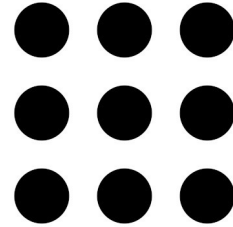
Testati tutti e 21 gli endpoint: output in immagini per ulteriore verifica manuale

Test overlay contorni



Ci/cd

- 1) **GitHub Actions**
- 2) **Test (PyTest)**
- 3) **Eseguibile (PyInstaller)**
- 4) **Documentazione Automatica (MkDocs)**
- 5) **Dependabot (GitHub Advisory Database)**



Limitazioni

- 1) Parametrizzazione: gestione/correzione non triviale
- 2) Risoluzione video in input: deve essere la stessa
- 3) Allineamento ROI: implica una corretta detezione
- 4) Fare più analisi (frontend)
- 5) [tempo]

